INCH-POUND

MIL-R-0010932P(GL)
17 August 1994
USED IN LIEU OF
MIL-R-10932N
30 June 1988

MILITARY SPECIFICATION

REFRIGERATORS, PREFABRICATED, PANEL TYPE

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-R-10932N. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This document covers prefabricated, walk-in-refrigerators assembled by the use of interchangeable panels. Refrigeration units for use with these refrigerators shall be panel-mounted (see 6.5).

1.2 Classification. Refrigerators shall be of the following sizes as specified (see 6.2).

Size
600 cubic feet
1200 cubic feet
1800 cubic feet
4000 cubic feet

1.2.1 Part identification number (PIN). The PIN shall consist of the basic number of the specification, and the appropriate PIN structure as specified in 6.4.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-WEE, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4110

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-344	- Lacquer (Clear Gloss)
A-A-665	- Lacquer
A-A-1927	- Padlock
L-P-320	- Pipe and Fittings, Plastic (Polyvinyl Chloride, PVC, Drain, Waste and Vent)
W-F-406	- Fittings for Cable, Power, Electrical and Conduit, Metal, Flexible
W-S-896/2	- Switches, Toggle and Lock, Flush Mounted (AC) (Detail Specification)
QQ-C-390	- Copper Alloy Castings (Including Cast Bar)
RR-C-271	- Chains and Attachments, Welded and Weldless
TT-S-176	- Sealer, Surface, Varnish Type, Floor, Wood and Cork
TT-V-121	- Varnish, Spar, Water-Resisting
MMM-A-121	- Adhesive, Bonding Vulcanized Synthetic Rubber To Steel

MILITARY

MIL-C-104	- Crates, Wood: Lumber and Plywood Sheathed, Nailed, and Bolted
MIL-P-116	- Preservation, Methods of
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank
MIL-T-704	- Treatment and Painting of Material
MIL-R-13312	- Refrigeration Units, Mechanical, Panel Mounted for Refrigerator, Prefabricated, Panel Type
MIL-V-13811	- Varnish, Waterproofing, Electrical, Ignition
MIL-F-14187	- Fastener, Refrigerator Panel
MIL-R-14328	- Rubber Sheet: Synthetic, Medium Soft, General-Purpose Gasket Material (For Extreme Climatic Conditions)
MIL-C-16173	- Corrosion Preventive Compound, Solvent Cutback, Cold-Application
MIL-I-46058	- Insulating Compound, Electrical (For Coating Printed Circuit Assemblies)

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MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-C-52950 - Crates, Wood, Open and Covered
MIL-C-53039 - Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of U.S. Military Property

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094, or from the Federal Supply Service Bureau, Specification Section (3FPB-W), Suite 8100, 470 East L'Enfant Plaza, SW., Washington, DC 20407.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

UNITED STATES DEPARTMENT OF COMMERCE

Product Standards PS 1 - For Construction and Industrial Plywood

(Copies may be obtained from the U.S. Department of Commerce, Technology Administration, National Institute of Standards and Technology, National Center for Standards and Certification Information, TRF Building, Room A163, Gaithersburg, MD 20899.)

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-13-1858 - Fastener, Refrigerator Panel; Assembly and Installation
5-13-2651 - Refrigerator, Panel Type, Prefabricated; Assemblies
5-13-2652 - Refrigerator, Panel Type, Prefabricated; Erection Instructions
5-13-2653 - Refrigerator, Panel Type, Prefabricated; Corner Panel "A"
5-13-2654 - Refrigerator, Panel Type, Prefabricated; Standard Wall Panel "B"
5-13-2655 - Refrigerator, Panel Type, Prefabricated; Walk-in Door Panel "C"
5-13-2656 - Refrigerator, Panel Type, Prefabricated; Walk-in Door
5-13-2657 - Refrigerator, Panel Type, Prefabricated; Evaporator Panel "D"

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5-13-2658 - Refrigerator, Panel Type, Prefabricated; Floor or Ceiling Panel "FL" & "CL"
5-13-2659 - Refrigerator, Panel Type, Prefabricated; Floor or Ceiling Panel "FR" & "CR"
5-13-2660 - Refrigerator, Panel Type, Prefabricated; Center Floor and Ceiling Panel "FC" & "CC"
5-13-2661 - Refrigerator, Panel Type, Prefabricated; Partition Panel "H"
5-13-2662 - Refrigerator, Panel Type, Prefabricated; Inside Drain
5-13-2663 - Refrigerator, Panel Type, Prefabricated; Outside Drain
5-13-2664 - Refrigerator, Panel Type Prefabricated; Gasket
5-13-2665 - Refrigerator, Panel Type, Prefabricated; Door Canopy
5-13-2666 - Refrigerator, Panel Type, Prefabricated; Ramp
5-13-2667 - Refrigerator, Panel Type, Prefabricated; Wrench, 5/16 Hexagonal
5-13-2668 - Refrigerator, Panel Type, Prefabricated; Typical Joint and Cutout Details
5-13-2669 - Refrigerator, Panel Type, Prefabricated; Floor Racks
5-13-2670 - Refrigerator, Panel Type, Prefabricated; Conveyor Door Panel "G"
5-13-2671 - Refrigerator, Panel Type, Prefabricated; Curtain Details

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: SATNC-WEE, Natick, MA 01760-5018.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

UNDERWRITERS LABORATORIES, INC. (UL)

UL 1 - Flexible Metal Conduit

UL 98 - Enclosed and Dead-Front Switches

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingston Road, Northbrook, IL 60062.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - The National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, One Batterymarch Park, Quincy, MA 02269.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 48	- Gray Iron Castings
A 576	- Steel Bars, Carbon, Hot-Wrought, Special Quality
A 675	- Steel Bars, Carbon, Hot-Wrought, Special Quality, Mechanical Properties
B 26	- Aluminum-Alloy Sand Castings
B 32	- Solder Metal
B 86	- Zinc-Alloy Die Castings
B 209	- Aluminum and Aluminum-Alloy Sheet and Plate
D 235	- Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent)
D 1974	- Methods of Closing, Sealing, and Reinforcing Fiberboard Shipping Containers
D 2559	- Adhesives for Structural Laminated Wood Products for Use Under Exterior (Wet Use) Exposure Conditions
D 2846	- Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic, Hot-and Cold-Water Distribution Systems
D 3950	- Strapping, Nonmetallic (and Joining Methods)
D 3953	- Strapping, Flat Steel and Seals
D 4442	- Direct Moisture Content Measurement of Wood and Wood Base Materials
D 4727	- Corrugated and Solid Fiberboard Sheet Stock (Container Grade) and Cut Shapes
D 5118	- Fabrication of Fiberboard Shipping Boxes
D 5330	- Pressure-Sensitive Tape for Packaging, Filament-Reinforced
D 5486	- Pressure-Sensitive Tape for Packaging, Box Closure, and Sealing
E 18	- Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN NATIONAL STANDARD/AMERICAN HARDBOARD ASSOCIATION (ANSI/AHA)

A135.4	- Basic Hardboard

(Application for copies should be addressed to the American Hardboard Association, 1210 W. Northwest Highway, Palatine, IL 60067.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2) a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Codes and standards. Refrigerator electrical components shall comply with the applicable requirements of Underwriters Laboratories (UL), Standards 1, and 98, and with NFPA Standard No. 70 (see 4.4.7).

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for refrigerators, provided the completed end item complies with all provisions of this document. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Cadmium, antimony, bismuth, lead, or other toxic materials shall not be used on any hardware or components in the interior of the refrigerator. Materials or material treatments which will potentially come into contact with foodstuffs and which could cause toxic reaction or transfer flavor or odor to foodstuffs shall not be used. Commercial tolerances shall apply to all metal thicknesses and diameters unless otherwise specified hereinafter or on drawings. Recycled or reclaimed materials may be used in the construction of the equipment described herein to the maximum extent possible. Recycled and reclaimed materials shall be defined as previously used material which has been reprocessed to become a source of new raw materials. Under no circumstances shall the contractor submit to the Government for acceptance, reconditioned or rebuilt components as part of the equipment described herein.

3.3.1 Lumber. Lumber used in the fabrication of components shall be hardwood or softwood, as specified herein. All lumber shall be kiln dried without honeycombing or case hardening. The moisture content at the time of fabrication shall not exceed 15 percent when tested as specified in 4.4.1.1.1. The lumber shall not be decayed, warped, or have shakes, clusters of knots greater than one-fourth of the width of the individual pieces of lumber, or surface cracks over 4 inches long. The slope of the grain of the lumber shall not exceed 1 inch in 10 inches when tested as specified in 4.4.1.1.2. Local deviation of the grain such as burl or birds eye shall be permissible. Lumber with open knot holes or loose knots shall not be acceptable. Sound, tight knots not more than 1-1/2 inch in largest diameter shall be allowable in lumber 3 inches or more in width. Sound, tight knots not more than 5/8 inch in largest diameter shall be allowable in parts less than 3 inches in width provided these knots are not positioned at the intersection of a joint or at the edge of a piece of lumber. Knots of maximum allowable size shall be not closer than 12 inches, center to center, in any one piece of lumber.

3.3.1.1 Hardwood. Hardwood shall be oak.

3.3.1.2 Softwood. Softwood shall be any of the following species: cypress, fir, hemlock, larch, redwood, spruce, or ponderosa, sugar, or Southern pine.

3.3.2 Plywood. Plywood shall be either Douglas fir plywood or Southern pine plywood grade A-B or better conforming to Product Standard PS 1.

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3.3.3 Wood preservative. Unless otherwise specified (see 6.2), wood preservative shall contain 99 percent dry cleaning fluid conforming to type II of ASTM D 235 and 1 percent solubilized copper-8-quinolinolate by weight. The solubilized copper of quinolinolate shall be free of amines, napthenic acid or its derivative.

3.3.4 Wood sealer and finishes.

3.3.4.1 Wood sealer. Wood sealer shall conform to class 2 of TT-S-176.

3.3.4.2 Varnish. Varnish shall conform to TT-V-121. Varnish prepared for spraying shall consist of adding not more than 1 pint of varnish makers' and painters' naphtha (VM and PN) to each gallon of varnish.

3.3.4.3 Lacquer. Two types of lacquer are required. One shall conform to A-A-665 and be clear, and the other shall conform to A-A-344.

3.3.5 Metal surface priming and coating.

3.3.5.1 Treatment and painting. The exterior areas of the refrigerator to be painted, shall be cleaned, treated and painted in accordance with type F or G of MIL-T-704. The color shall be as specified in 3.3.5.2. The word "CARC" with the month and year of coating application, shall be stencilled in 1 inch high block letters using lusterless Black 37030, type III or IV, conforming to MIL-C-46168 or Black conforming to MIL-C-53039 and shall be located near the identification plate.

3.3.5.2 Finish coat for metal surfaces. The primed exterior metal surfaces shall be painted Camouflage Green 383, 34094 of MIL-C-46168, or Camouflage Green 383, 34094 of MIL-C-53039.

3.3.6 Insulation. Insulation shall be foamed-in-place polyurethane material and shall have a manufacturers rated K factor of not more than 0.15. Insulation shall have a free rise density of not less than 1.7 pounds per cubic foot when tested as specified in 4.4.2.1.1 or an in-place density of not less than 2 pounds per cubic foot when tested as specified in 4.4.2.1.2. The foamed-in-place insulation shall be dimensionally stable at a temperature of 150°F when tested as specified in 4.5.2.

3.3.7 Adhesive. The wood joining adhesive shall conform to ASTM D 2559 or equivalent, and shall be compatible for food storage use. The adhesive shall be waterproof and water-resistant, and fungus and rodent resistant. The adhesive shall contain no toxic or formaldehyde type components. The manufacturer shall provide to the contracting officer the type that he proposes to use for approval prior to fabrication (see 3.3).

3.3.8 Gaskets.

3.3.8.1 Gaskets; door and panels. The gasket as used on the doors and panels, except partition panels, shall be fabricated of flexible urethane core enclosed in a flexible urethane coating, with a

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reinforced single seal and of the dimensions as shown on Drawing 5-13-2664. Ends shall be joined using adhesive as specified in MMM-A-121.

3.3.8.2 Gaskets (partitions). The gaskets used on the sides and bottom of partition panels shall be synthetic rubber. The gaskets used on the top of partition panels shall be synthetic heavy duty sponge rubber and have a thin synthetic rubber coating on four sides resistant to deterioration from oils, fats, moisture, and direct solar radiation and shall conform to the thicknesses as shown on applicable drawings.

3.3.9 Aluminum alloy plate and sheet. Aluminum alloy plate and sheet shall be 5052 and conform to temper H34 of ASTM B 209.

3.3.10 Steel. Steel bar shall conform to ASTM A 576 and ASTM A 675.

3.3.11 Castings.

3.3.11.1 Iron castings. Iron castings shall conform to class 25 of ASTM A 48.

3.3.11.2 Aluminum castings. Aluminum castings shall conform to alloy 514, temper F of ASTM B 26.

3.3.11.3 Bronze castings. Bronze castings shall conform to alloy C92000, C82400, C82500, C82600, C82700, C82800, type I of QQ-C-390. The finished castings shall be given a chromium, nickel, or zinc finish.

3.3.11.4 Die-cast zinc. When die-cast zinc is used the zinc-base alloy casting shall conform to either alloys AG 40A or AC 41A of ASTM B 86.

3.3.12 Nonmetallic strip and tubing, rubber and plastic.

3.3.12.1 Building board. Material for breaker strips shall conform to class 2, (S1S) of ANSI/AHA A135.4, and the surface shall be unfinished plain board.

3.3.12.2 Nonmetallic tubing. The tubing shown on Drawing 5-13-2656 used to shield the door latch plunger from the insulation shall be nonmetallic, nonflexible, waterproof material with a low heat conductivity and shall conform to ASTM D 2846.

3.3.12.3 Rubber, sheet and strip. The drain plug gasket and outside drain adapter gaskets shall be synthetic rubber in the thicknesses shown on the applicable drawings, and conform to MIL-R-14328.

3.3.12.4 Pipe and tube. Plastic pipe and tubing used for floor drains in panels "FL" or "FR" shall be rigid plastic and conform to L-P-320. The plastic shall be nontoxic, resistant to acids, alkalies, metallic salts, and deterioration.

3.3.13 Solder. Solder shall conform to alloy composition A.1.1.1, Sn96 or A.1.1.2 Sn95, Sn94 E, AC and AM of ASTM B 32.

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3.3.14 Electrical components. Wire shall be installed within flexible conduit in accordance with NFPA Standard No. 70. Wiring shall consist of two No. 14 AWG conductors and a ground wire. Conduit shall conform to UL Standard 1.

3.3.14.1 Fittings, conduit. Fittings shall conform to type I, class 4, style N of W-F-406.

3.3.14.2 Switch. Switches shall conform to W-S-896/2-03-A. The switch shall be housed in a vapor-tight enclosure conforming to UL Standard 98, and shall be externally operated and mounted in the cover of the box inclosure.

3.3.14.3 Pilot light. The pilot light shall be 6-watt, 120-volt incandescent lamp having a candelabra screw base, commercial type S-6, and enclosed in a vaportight housing. The lamp shall be supported within the enclosure and wired to illuminate a red glass jewel indicator in the cover of the box enclosure when the switch is in the "on" position.

3.3.14.4 Lighting fixture. The lighting fixture to be furnished shall be a vapor proof, bracket type, lamp fixture, and shall be mounted as shown on Drawing 5-13-2655. The vapor proof lamp fixture shall be equipped with a screw type shatter-proof globe of the polycarbonate type plastic, and shall be complete with a 100-watt, 120-volt, inside-frosted incandescent bulb having a medium screw base.

3.3.14.5 Plug and receptacle. The plug and plug receptacle shall be provided and shall be of the weather tight type, designed for 3-wire, 2-pole, grounded-service and shall be rated at 15 amperes, 125 volts. The receptacle shall be of the recessed male, locking type. A cap or flap cover shall be provided to seal the receptacle housing when not in use and shall be secured to the housing to prevent loss. The plug shall contain a sealing ring to form a weather tight joint when the plug is inserted in the receptacle. A warning decal shall be placed adjacent to the receptacle and shall state "DANGER - THIS RECEPTACLE MUST BE CONNECTED TO A GROUNDED SERVICE".

3.3.15 Hardware.

3.3.15.1 Door latch and strike. The walk-in door and conveyor door latch and strike assembly shall be of the commercial, adjustable type and of high strength aluminum alloy, zinc die cast conforming to 3.3.11.4, or of cast or forged brass, or malleable iron with a finish compatible for CARC primer and paint application. The latch shall have provisions for holding the padlock specified in 3.3.15.5. When a die-cast latch and strike in furnished, the handle shall be forged brass or malleable iron and the striker tongue shall be extruded brass or forged steel. The strike shall be so constructed that removing the bolt or screw from the strike housing shall not allow the locked door to be opened. The walk-in door shall be provided with an inside release handle for exit during normal operation. The walk-in door latch construction shall permit a person trapped in the refrigerator to escape when the refrigerator is padlocked on the outside. The

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walk-in door latch and inside release mechanism shall be so designed and constructed that if the door is opened from the inside when padlocked on the outside, no component part shall become distorted, dismounted, or detached.

3.3.15.2 Hinges, walk-in and conveyor door. Hinges for walk-in door shall be of the commercial strap type or self-closing type matching the latch of the door in appearance. Hinges for conveyor door shall be of the commercial strap type. The commercial strap type hinge for walk-in door shall be not less than 12 inches long, for self closing type not less than 10 inches long, and for conveyor door not less than 9 inches long. The hinges shall be constructed of forged brass, malleable iron, steel or zinc die cast conforming to 3.3.11.4 with a finish compatible for CARC primer and paint application. The strap type hinges for the walk-in door and conveyor door shall have ball bearing or self-lubricating type washers and shall be of the fixed steel pin type with the unsupported section the strongest part of the hinge. The self-closing type hinges shall have lubrication accomplished through the use of self-lubricating nylon cam insert.

3.3.15.3 Panel fasteners. Unless otherwise specified (see 6.2), panel fasteners shall conform to MIL-F-14187.

3.3.15.4 Wrench for panel fasteners. Hexagonal wrenches, 5/16-inch size, shall be fabricated according to Drawing 5-13-2667 from steel specified in 3.3.10 and heat treated to a Rockwell C hardness of 36-44 (see 4.4.1.1.3). Two wrenches shall be attached to each evaporator panel "D" as shown on Drawing 5-13-2657. For Marine Corps procurements (see 6.2), three wrenches shall be furnished for each "D" panel.

3.3.15.5 Padlock. Unless otherwise specified (see 6.2), the padlock shall conform to A-A-1927. The padlock shall be securely attached to the walk-in and conveyor door by chain as supplied.

3.3.15.6 Bolts, nuts, washers, screws, street elbows, and barrel bolts. Unless otherwise specified on the drawings, bolts, nuts, washers, screws, street elbows, and barrel bolts shall be zinc coated steel or chrome plated brass. Brass or corrosion-resisting steel screws shall be used in the fabrication of the floor racks.

3.3.15.7 Nails, steel, common. Nails shall be aluminum, cement or zinc coated, and in the sizes shown on applicable drawings.

3.3.15.8 Chain, safety. The safety chain used for the drain plug retainers shall conform to type II, class 6 of RR-C-271 and shall be plated to resist corrosion or may be of commercial grade corrosion resistant steel.

3.3.15.9 Thermometer. A thermometer shall be provided (see 3.4.5) and shall read to -40°F to 120°F, with an allowable exposure temperature to 155°F. The case shall be corrosion-resistant, provided with two screw mounting holes, and be at least 3 inches wide and 14 inches high.

3.3.16 Tape. Tape shall conform to type I or IV, 4 inch width, color Dark Green, number 34108 of ASTM D 5486.

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3.4 Design and construction. Refrigerators shall be 12 feet, 9-5/8 inches in overall width and shall be of 600, 1200, 1800, and 4000 cubic-foot sizes. Refrigerators and component parts thereof shall be as specified herein and on the drawings listed in 2.1.1. Unless otherwise specified (see 6.2), refrigerators shall consist of panels designated in the panel schedule on Drawing 5-13-2651 to accommodate the refrigeration unit covered by MIL-R-13312 (see 6.5). All exterior panel skins shall be painted as specified (see 3.3.5.1 and 3.3.5.2). The panels shall assemble to each other and no special tools other than furnished shall be necessary to assemble or disassemble all panels when tested as specified in 4.5.3. The tape specified in 3.3.16 shall be used for sealing ceiling panels and perimeter joints between wall and ceiling panels. One roll of tape shall be furnished with each size 600 and 1200 cubic feet, two rolls shall be furnished with each size 1800 cubic feet and four rolls furnished with each size 4000 cubic feet.

3.4.1 Heat leakage. The heat leakage of a complete 600 cubic foot refrigerator shall be no greater than 0.070 British thermal units (Btu's) per square foot, per degree F. The outside area of the refrigerator shall be used for heat leakage calculations (see 4.5.1).

3.4.2 Panel construction.

3.4.2.1 Panel skins. The metal skins shall be made from aluminum alloy specified in 3.3.9 and shall have a minimum thickness of 0.050 inch. Solder specified in 3.3.13 shall be used in soldering skins. All seams except as otherwise stated shall be soldered in accordance with 3.7.2 or welded in accordance with 3.7.1. All seams shall be tight to effect a positive moisture vapor seal.

3.4.2.2 Panel sheets, number per skin. The panel skins for panels C, D, and G shall be fabricated from not more than four sheets of metal per skin. The panel skins for partition-panels H-1, H-2, and H-3 shall be fabricated from not more than two sheets of metal per skin. All other panel skins shall be fabricated from one sheet of metal. Sheets notched or slit to accommodate forming shall be fitted where necessary with metal filler pieces of the same type and thickness as the skins and then soldered or welded into place with the exception of the horizontal 3/4-inch corner flange seams on the floor, ceiling and corner panels and other specific seams as noted on applicable drawings.

3.4.2.3 Wood members. Wood (lumber and plywood) members shall be as specified in 3.3.1 and 3.3.2. All solid wood and plywood members shall be of one-piece construction when fabricated in their final form. Scarfing of plywood members in accordance with American Plywood Association requirements and finger jointing of solid wood are acceptable. All solid wood members, except door jambs, sills, and floor racks, shall be fabricated from softwood specified in 3.3.1.2. Door jambs, sills, and floor racks shall be fabricated from hardwood specified in 3.3.1.1. All wood members except oak, plywood sheets on floor and ceiling panels, and varnished or lacquered members used in the fabrication of the refrigerator shall be immersed in the preservative specified in 3.3.3 for not less than 15 minutes and air

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dried until free of solvent. Necessary trimming or resurfacing of complete preservative treated wood members shall be permitted to secure the required fit provided bare surfaces are thoroughly coated, prior to assembly, with wood preservative specified herein.

3.4.2.4 Adhesive cementing. Wherever solid wood jambs, sills, or members used to form tongues and grooves are joined to the plywood or solid wood members, adhesive specified in 3.3.7 shall be used in addition to the application of screws or nails as shown on applicable drawings. Where it is specified on drawings for the metal skins to be cemented to plywood backing, the skins shall be bonded to the plywood with a permanent waterproof cement of a commercial grade suitable for bonding metal to wood. Preparation and application of adhesives and cement shall be done in a uniform manner with quantities recommended by the manufacturer. Assembly periods, clamping and pressing time, and pressure drying time shall be in accordance with manufacturer's recommendations.

3.4.3 Insulation. Insulation shall be a non-moisture absorbing material; shall be homogeneous throughout; and shall have "K" factors as specified below using a 70 F mean temperature and expressed in terms of British thermal units (Btu) per hour per square foot per inch of thickness per degree F temperature difference. Loose fill type of insulation will not be acceptable. Insulation shall be of the type that is normally used by the manufacturer.

3.4.3.1 Insulation installation. All space within each panel shall be completely filled with foamed-in-place insulation as shown on panel drawings (see 2.1) and specified in 3.3.6. Prior to foaming, the panel inner and outer liner shall be treated in accordance with the manufacturer's standard commercial practice to insure permanent bonding of the insulation to the liner material.

3.4.4 Panels.

3.4.4.1 Conveyor door panel (G) and partition panels (H-1, H-2, and H-3). When specified (see 6.2), conveyor door panels shall be furnished in the number specified (see 6.2), and shall replace standard wall panels (B) or walk-in door panels (C), as specified (see 6.2). When specified (see 6.2), partition panels shall be omitted.

3.4.4.2 Walk-in door panel. Each walk-in door panel shall be complete with door attached. Unless otherwise specified (see 6.2), the door shall be hinged on the right hand side when viewed from outside the refrigerator.

3.4.5 Thermometer attachment. All walk-in door panels shall have a thermometer specified in 3.3.15.9 attached in the locations shown on Drawing 5-13-2655 with zinc-coated sheet metal screws of suitable size.

3.4.6 Ramp and door canopy. The door canopy and, when specified (see 6.2), the ramp shall be furnished as shown on Drawings 5-13-2665 and 5-13-2666. One each shall be required with each walk-in door panel.

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3.4.7 Finish.

3.4.7.1 Metal surfaces. All external surfaces (see 3.4) shall be finished as specified in 3.3.5.

3.4.7.1.1 Cleaning, conditioning, and surface preparation. All exposed outer panel skins shall be thoroughly cleaned conditioned and prepared for painting as specified in 3.3.5.1.

3.4.7.1.2 Priming. All metal surfaces prepared for painting shall receive a coat of primer specified in 3.3.5.1.

3.4.7.1.3 Finish coat. All exterior metal surfaces to be painted shall be thoroughly coated with camouflage paint as specified in 3.3.5.2. The paint shall level out to a smooth and uniform film free from runs, wrinkles, drops and areas of thin film or no film.

3.4.7.2 Wood surfaces. The exterior surface of the jambs and sills of the walk-in door panels, all exposed wood surfaces of the walk-in doors, conveyor doors and openings, and refrigeration unit openings shall be given one coat of wood sealer specified in 3.3.4.1 followed by two coats of spray varnish specified in 3.3.4.2 or lacquer conforming to A-A-344 as specified in 3.3.4.3.

3.4.7.3 Floor racks. Floor racks, after fabrication, shall be given one coat of wood sealer specified in 3.3.4.1 followed by two coats of spar varnish specified in 3.3.4.2 or lacquer conforming to A-A-344 as specified in 3.3.4.3.

3.4.8 Breaker strips. The breaker strips shall be constructed of the material specified in 3.3.12.1, and shall be removable. All surfaces shall be given one coat of sealer as specified in 3.3.4.1 followed by two coats of varnish specified in 3.3.4.2, or lacquer specified in 3.3.4.3 prior to installation.

3.5 Erection instruction.

3.5.1 Erection instruction, decalcomanias. Standard commercial decalcomanias illustrating detail erection instructions as shown on Drawing 5-13-2652 shall be affixed to each walk-in door as shown on Drawing 5-13-2656. The decalcomania shall be approximately 11 1/2 inches high and shall be overcoated after drying with a clear lacquer conforming to A-A-665 as specified in 3.3.4.3.

3.5.2 Erection instructions, printed. Erection instructions specified in 3.5.1 shall be furnished printed on a good commercial grade of bond paper and supplied as specified in 5.1.1.4.1.

3.6 Marking for identification. All equipment assemblies, and parts shall be marked for identification in accordance with MIL-STD-130, except as specified hereinafter.

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3.6.1 Panel identification. Each panel shall be identified with the letter designation as shown on applicable drawings. The word "TOP" shall also be marked on the following panels: corner panel A, standard wall panel B, evaporator panel D, and conveyor door panel G. These markings shall be stenciled or printed with waterproof ink and in characters 1 inch high.

3.6.2 Data name plate. the refrigerator shall have a metal identification plate attached to the outside of the walk-in door as shown on Drawing 5-13-2656. The metal plate shall conform to type III, composition A, class 1 or 2 of MIL-P-514 and shall be permanently attached with screws or rivets. The identification plate shall be marked in accordance with MIL-STD-130 as follows:

1. U.S.
2. Military model number and nomenclature.
3. National Stock Number (NSN).
4. Serial number.
5. Contract number.
6. Manufacturer's name and address.
7. Model number.
8. CAGE number.
9. Technical manual number.

3.6.2.1 Data plate. For Marine Corps acquisitions (see 6.2), a data plate shall be attached to each refrigerator. These plates may be obtained from the Commandant of the Marine Corps (Code CS6), Headquarters Marine Corps, Washington, DC 20380. Data applicable to blank spaces 1, 3, 8, 9, and 10 shall be furnished with the plates. The contractor is required to imprint the data in all blank spaces and permanently affix the plate to the refrigerator.

3.7 Workmanship. All components and assemblies of the refrigerator shall be free from dirt and other extraneous material, burrs, slivers, rough die, tool, and grind marks, dents and cracks. Castings, molded parts and stampings, if used, shall be free of sand, fins, pits, blow holes and sprues. External surfaces shall be free from sharp edges and corners, except where corners are required.

3.7.1 Welding. All welding shall be as shown on applicable drawings. Materials to be welded shall be clean and free of rust, scale, corrosion, oil, water, and any other matter injurious to welding. Welds shall be sound and smooth, free from spatter, and show good fusion. Welds shall have equivalent strength of sections joined. All scale or flux deposits shall be removed from the finished welds.

3.7.2 Soldering. Seams to be soldered shall be completely filled and wiped smooth after soldering. Flux shall not be retained by the seam after soldering is completed. Residual flux shall be inhibited by a hot water rinse. Solder shall not be used on surfaces of equipment with which unpackaged, edible food may come in direct contact during normal use of the equipment. Welding shall be acceptable wherever soldering is specified.

3.7.3 Metal fabrication. Metal used in the fabrication of the refrigerator shall provide for original quality surface finish and shall be free of kinks and unspecified bends. Forming and shearing

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shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bonds of a major nature shall be made with dies or fixtures to insure uniformity of size and shape.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and shall be tested for the characteristics specified in 4.5.1, 4.5.2, and 4.5.3. The presence of any defect, any dimension not within specified tolerances, or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component testing.

4.4.1.1.1 Moisture content. Lumber shall be tested for moisture content in accordance with ASTM D 4442 to determine compliance with 3.3.1. The sample unit shall be one piece of lumber. Three determinations shall be made on each sample unit and the results shall be the average of these determinations. The inspection level shall be S-1 and the finding of any defect shall be cause for rejection of the lot. The lot shall be all pieces of lumber offered for inspection at one time.

4.4.1.1.2 Slope of grain. The slope of grain shall be measured to determine compliance with 3.3.1. It may be determined by actual measurement with a straight edge and ruler. The lot shall be all pieces of lumber offered for inspection at one time. The sample unit shall be one piece of lumber. The inspection level shall be S-1 and the finding of any defect shall be cause for rejection of the lot.

4.4.1.1.3 Hardness. The lot of wrenches shall be tested for hardness in accordance with ASTM E 18 to determine compliance with 3.3.15.4. The sample unit shall be one wrench. The inspection level shall be S-4 and the finding of any defect shall be cause for rejection of the lot.

4.4.1.1.4 Thermometer. To determine compliance with 3.3.15.9, the thermometer shall be placed in a minimum ambient temperature of 155°F for 1 hour. The thermometer shall then be observed for damage and checked against a calibrated thermometer to assure that it reads properly. The lot size shall be expressed in units of thermometers. The sample unit shall be one thermometer. The inspection level shall be S-1 and the finding of any defect shall be cause for the rejection of the lot.

4.4.2 In-process inspection. Inspection shall be made of the following operations and requirements to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and to the operations.

a. Wood preservation (see 3.4.2.3)
b. Correct selection and construction of lumber (see 3.4.2.3 and applicable drawings)
c. Adhesive cementing (see 3.4.2.4)
d. Installation of insulation (see 3.4.3)
e. Liner treatment prior to foaming (see 3.4.3.1)
f. Cleaning and conditioning of metal surfaces (see 3.4.7.1.1)
g. Priming metal surfaces (see 3.4.7.1.2)
h. Finishing of metal surfaces (see 3.4.7.1.3)
i. Sealing, varnishing (see 3.4.7.2 and 3.4.7.3)
j. Breaker strips treated (see 3.4.8)
k. Welding (see 3.7.1)
l. Soldering (see 3.7.2)

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4.4.2.1 In-process testing.

4.4.2.1.1 Free rise density test for insulation. The sample shall be a 6-inch cube cut from the center of a minimum 8-inch free foam cube of the insulation. The 6-inch cube shall be weighed and the density shall be calculated as follows:

$$\text{Density (lbs./ft}^3\text{)} = \text{Weight of 6-inch cube (lbs.)} \times 8$$

Any nonconformance with the above or with the density requirement of 3.3.6 shall constitute failure of this test and rejection of the lot.

4.4.2.1.2 Foamed-in-place density test for insulation. Prior to the beginning of each day's production of refrigeration panels, one sample of the insulation to be used in that day's production shall be tested as specified in 4.4.2.1.2.1 for conformance to the applicable density requirements as specified in 3.4.3 and 3.4.3.1.

4.4.2.1.2.1 Foamed-in-place test. The weight and interior volume of a refrigerator panel without foam insulation shall be determined. The panel shall then be foamed and weighed. The foam density shall then be calculated as follows:

$$\text{Density (lbs./ft}^3\text{)} = \frac{\text{Panel weight (lbs.) after foaming} - \text{Panel weight (lbs.) before foaming}}{\text{Panel interior volume}}$$

Any nonconformance with the above or with the density requirement of 3.3.6 shall constitute failure of this test and rejection of the lot.

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of refrigerators. The sample unit shall be one refrigerator. The inspection level shall be II and the finding of any defect shall be cause for rejection of the lot.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish 201	Color not as specified		
Construction and workmanship 202	Component missing or not specified type		101
	Component not in proper location		
Panels 203	Soldered seams not smooth		
	More sheets per skin than specified		

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204		
	Corner joints, cracks, and crevices not sealed	102
	Filler pieces not as specified	
205		
Marking		
Data name plate	Missing, incomplete, not as specified	103
Instruction	Missing, incomplete, not as specified	104
Identification	Missing, incomplete, not as specified	
206		

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimensions not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of refrigerators. The sample unit shall be one refrigerator. The inspection level shall be S-2 and the finding of any defect shall be cause for the rejection of the lot.

4.4.5 End item testing. When a first article is not required, the first production unit shall be tested as specified in 4.5.1, 4.5.2, and 4.5.3. Any nonconformance shall be cause for rejection. The test specified in 4.5.3 shall be performed on the first complete refrigerator of each size which shall also be considered the first production unit, and on every tenth refrigerator. Any failure of the first production unit shall be cause for rejection of the unit. Any nonconformance of a tenth tested refrigerator shall be cause for rejection of that unit and the previous five units shall be tested. Defective panels may be corrected and resubmitted for test. If there is no indication of failure during these five tests, the frequency of testing will revert to every tenth refrigerator. If there is failure within the five tests, the complete lot shall be rejected.

4.4.6 Packaging examination. The fully packaged items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the finding of any defect shall be cause for the rejection of the lot.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing, damaged, or not as specified.

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Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling. Bulged or distorted container.
Content	Number per container is more or less than required.

4.4.7 Code and standard compliance. Certifications, certified test reports, or listing marks for standards, submitted in accordance with 3.2, shall be examined and validated as proof of compliance. Prior to Government approval of first shipment, the contractor shall submit to the contracting officer or his authorized representative for approval, certification that the refrigerators he proposes to furnish, meet the applicable requirements of UL and NFPA as follows.

4.4.7.1 Underwriters Laboratories, Inc., (UL). Acceptable evidence of meeting the requirements of the Underwriters Laboratories standards 1 and 98 (see 3.2), shall be a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the electrical components have been tested and conform to the US Standards for Safety.

4.4.7.2 National Fire Protection Association (NFPA). Acceptable evidence of meeting the requirement of NFPA shall be the manufacturer's certified statement that the electrical components conform to the applicable requirements of NFPA No.70.

4.5 Methods of inspection. When a first article is required, the test specified in 4.5.1 may be performed on a 600 cubic-foot size refrigerator regardless of the size specified in the contract. When the test specified in 4.5.1 is to be given to the first production unit, the test unit may be a 600 cubic-foot size refrigerator regardless of the size specified in the contract.

4.5.1 Heat leakage test. A heat leakage test shall be conducted with the refrigerator effected in a controlled temperature ambient and a monitored heat input supplied to the refrigerator to determine compliance with 3.4.1. All temperature, and heat input shall be recorded.

$$\text{Heat leakage} = \frac{\text{Total watt-hours, input X Btu/watt-hour}}{\text{Length of test (hours) X Temperature difference (°F) X Total outside area of refrigerator (square feet)}}$$

The temperature difference shall be not less than 60°F nor more than 80°F. The ambient temperature shall be the average of eight readings taken at points located 6 inches from each outside corner of the box. The internal, or box temperature, shall be the average of eight readings taken at points located 6 inches from the inner walls at each corner of the box. Eight ambient and box temperature readings shall be taken at consecutive one-half hour intervals with ambient temperature variations not more than plus or minus 2°F and box temperature variations not more than plus or minus 1°F. At least one interior and one exterior temperature reading shall be taken on a recording instrument. The refrigerator shall

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be erected on three dunnage rails to provide a minimum air space of 4 inches between the floor and the bottom of the refrigerator. The refrigerator shall be no closer than 12 inches from any wall or ceiling. Heat leakage in excess of that specified in 3.4.1 shall constitute failure of the test.

4.5.2 Heat stability test. One each of panels A, B, C, D and FC from the first production refrigerator shall be dimensionally measured for wall thickness and placed in an oven or heated room and held at 150°F for 24 hours. After removal of the panels from the oven, any indication of swelling shall require a measurement to determine if the foam had a swell of more than 5 percent to determine conformance to 3.3.6. Inability of the insulation to remain stable within 5 percent shall constitute failure of the test.

4.5.3 Assembly test. The panels for one complete refrigerator shall be assembled and disassembled twice to determine compliance with 3.4. Panels shall be interchanged for the second assembly. Inability of the panels to assemble in accordance with 3.4 shall constitute failure of the test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Disassembly. The thermometer, light bulb and globe shall be detached from the panel.

5.1.1.2 Repainting and recoating. Painted and permanently coated surfaces, from which the paint or permanent coating has been removed or which are defective, shall be repainted or recoated with material identical with the original.

5.1.1.3 Preservative application. Preservative application shall be in accordance with material and method requirements of MIL-P-116 as hereinafter specified.

5.1.1.3.1 Panel fastener wrenches. Each panel fastener wrench shall be thoroughly cleaned with solvent conforming to ASTM D 235. After being thoroughly dried, each wrench shall be completely coated with rust preventive compound type I of MIL-C-16173, and secured in place on the "D" panel as shown on Drawing 5-13-2657 with tape conforming to type I or IV of ASTM D 5486 (see 3.3.16).

5.1.1.3.2 Switches. Each switch shall be sealed or masked with tape specified in 5.1.1.3.1. The exposed surfaces of the tape shall be coated with varnish conforming to MIL-V-13811.

5.1.1.4 Unit packing. Packing shall be in accordance with material and method requirements of MIL-P-116 as hereinafter specified.

5.1.1.4.1 Thermometer, globe, incandescent bulb, plug, tape, and erection instructions. One thermometer and attaching hardware shall be unit packed conforming to method III. One globe, incandescent bulb, plug, and tape shall be unit packed conforming to method III. The erection instruction sheet specified in 3.5.2 shall be unit packed conforming to method IC-3. The unit packed thermometer together with the unit packed

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erection instructions, globe, incandescent bulb, plug, and tape shall be further packed in a snug-fitting box conforming to style RSC, grade V3c, V3s, or V4s of ASTM D 5118. Each fiberboard box shall be closed in accordance with ASTM D 1974.

5.1.1.4.2 Panels. Panels C and G shall have strips of barrier-material conforming to type I, grade A of MIL-B-121 inserted between the door frame and the door gaskets and the door secured.

5.1.1.4.3 Conveyor door panel. The rubber curtain shall be secured in place by means of two pieces of fiberboard conforming to V3s of ASTM D 4727. The fiberboard shall be cut to the same height and width as the rubber curtain; one piece placed on each side of the curtain and the two pieces secured together with tape conforming to type I or IV of ASTM D 5486, or type IV of ASTM D 5330. Care shall be exercised so that the rubber strips are in their normal position prior to taping the fiberboard. The fiberboard enclosed curtain shall then be secured from movement with three encircling bands of minimum 3/8 by 0.015-inch metal or nonmetallic strapping conforming to type I, finish A or B of ASTM D 3953; or type II of ASTM D 3950, respectively. The padlock shall be locked in place on the door latch. The keys shall be wire-tied to the lock with minimum 0.057 inch diameter annealed wire, and further secured to the lock with tape specified herein. The lock shall be secured to the door latch with tape specified or other means to prevent movement while in transit.

5.2 Packing. Packing shall be level A.

5.2.1 Level A packing.

5.2.1.1 Components of refrigerators. Components of refrigerators, preserved as specified in 5.1, shall be packed on edge in the most compact manner and in combination and multiples as specified in 5.2.1.1.1 and 5.2.1.1.2, respectively, within nailed wood crates. Unless otherwise specified herein, crates shall conform to type V, style A, class 1 of MIL-C-52950, except that the length dimension shall be waived, with contents anchored and waterproofed in accordance with the appendix thereto. When specified (see 6.2), crates shall conform to type 1, class 2, style a of MIL-C-104, modified to fasten the end and side panels to the base of the crate by means of lag screws or nut-sleeve assemblies in lieu of nails, with contents anchored in accordance with the appendix thereto. Blocking and bracing shall not bear against the panels at points that are susceptible to puncture, damage by distortion, or shock in transit. Protruding hardware and gaskets shall be protected and shall not contact the inner surfaces of the crate, or the blocking and bracing. Care shall be exercised to prevent damage to gaskets attached to refrigerator panels.

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5.2.1.1.1 Pack designation and contents when components of each refrigerator are procured by size designation.

Size	Pack designation and number required
600	A, B, and C
1200	A, B, C, D, and H
1800	A, B, C, D, E, and 2 of H
4000	A, B, C, D, E, and 2 of F, 2 of G, and 6 of H

5.2.1.1.1.1 Contents of packs for refrigerator.

Pack A

3 Std wall panels 1/
1 Walk-in door panel with door 1/
1 Evaporator panel
4 Corner panels
1 Canopy
1 Ramp, when required
1 Fiberboard box containing globe, incandescent bulb, plug, erection instructions, tape and thermometer (see 5.1.1.4.1).

Pack B

5 Std wall panels 1/ - B

Pack C

1 Ceiling panel - CL
1 Ceiling panel - CE
1 Floor or ceiling panel - FL
1 Floor or ceiling panel - FR
1 Center ceiling panel - CC
1 Center floor panel - FC
4 Floor racks 24-1/4 inches wide
2 Floor racks 41-3/4 inches wide

Pack D

3 Std wall panels 1/ - D
1 Evaporator panel - D

Pack E

3 Std wall panels 1/ - B
1 Walk-in door panel with door 1/ - C
1 Partition each of H-1, H-2, and H-3 2/ - H
1 Canopy
1 Ramp, when required
1 Fiberboard box containing globe, incandescent bulb, plug, erection instructions, tape and thermometer (see 5.1.1.4.1)

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Pack F

1 Walk-in door panel with door 1/	- C
1 Evaporator panel	- D
1 Partition each of H-1, H-2, and H-3 2/	- H
1 Canopy	
1 Ramp, when required	
1 Fiberboard box containing globe, incandescent bulb, plug, erection instructions, tape and thermometer (see 5.1.1.4.1)	

Pack G

6 Std wall panels 1/	- B

Pack H

2 Center ceiling panels	- CC
2 Center floor panels	- FC
4 Floor racks 41-3/4 inches wide	

1/ When one or more conveyor door panels (G) are required, they shall replace a standard wall panel (B) or walk-in door panel with door (C).

2/ When partitions (H-1, H-2, H-3) are not required, they shall be omitted from the appropriate pack.

5.2.1.1.2 Components of refrigerator when procured in panel forms and when panels are procured for replacement. The components preserved as specified in 5.1, shall be packed as specified in 5.2.1.1.

5.2.1.1.2.1 Corner panels. Twenty-four panels shall be nested in fours and then strapped with two steel straps, 5/8 by 0.020 inch conforming to ASTM D 3953. The straps shall be placed approximately 7 inches from each end. Metal protectors shall be provided under the straps at the corners to prevent damage to the panels. Six bundles shall then be packed per crate.

5.2.1.1.2.2 Standard wall panels-B and evaporator panel D. Eight panels of one type only, shall be packed per crate.

5.2.1.1.2.3 Walk-in door panels with doors, -C canopies, ramps; and fiberboard boxes. Four walk-in door panels with doors, four canopies, four ramps when required, and four fiberboard boxes containing the components listed in 5.1.1.4.1, shall be packed per crate.

5.2.1.1.2.4 Conveyor door panels-G. Four conveyor door panels shall be packed per crate. The doors shall be positioned in the crate flat, with the exterior surface of the doors facing up.

5.2.1.1.2.5 Floor or ceiling panels-FL, CL, FR, and CR. Six floor or ceiling panels, of one type only, shall be packed per crate.

5.2.1.1.2.6 Center floor or ceiling panels-FC or -CC. Five center floor or ceiling panels shall be packed per crate.

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5.2.1.1.2.7 Partitions -H (H-1, H-2 and H-3). Twelve partition panels (four each of H-1, H-2, and H-3), shall be packed per crate

5.2.1.1.2.8 Floor racks. Twenty-four floor racks, of one width only, shall be packed per crate.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Additional marking. The outside of each erection instruction sheet package shall be stenciled as follows:

"ERECTION INSTRUCTIONS"

5.3.1.1 Crate marking. In addition to marking specified herein, each crate containing pack A shall have legibly printed or stenciled on the sides or ends, in characters not less than 2 inches in height, the following:

"THIS CRATE CONTAINS ERECTION INSTRUCTIONS"

5.3.1.2 Set marking. In addition to the marking specified herein, set markings shall be applied.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The prefabricated refrigerators are intended for use as self-supporting, sheltered or unsheltered refrigerators for the preservation of foodstuffs or other temperature-sensitive items.

6.2 Acquisition requirements. Acquisition documents must specify the following:

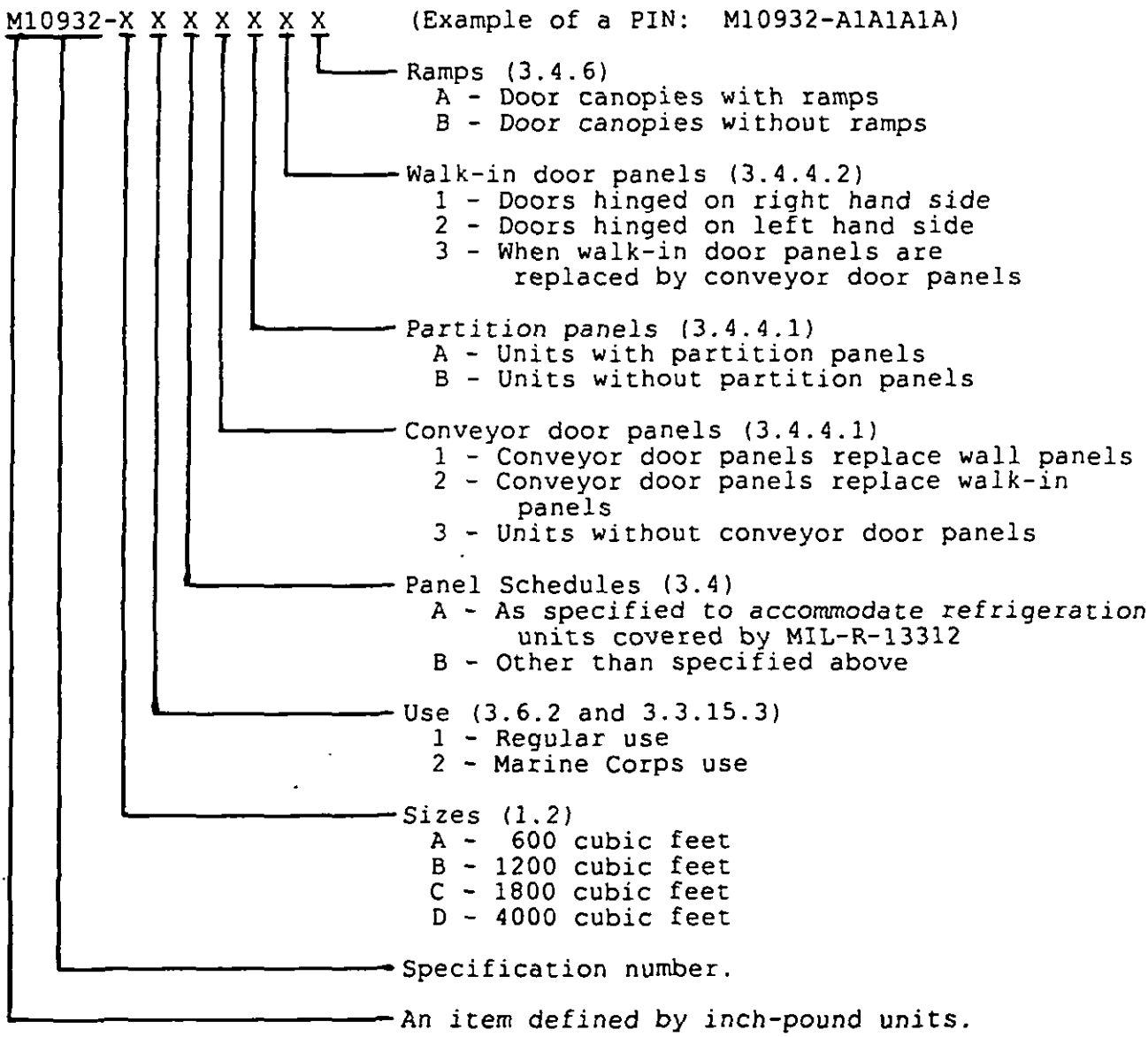
a. Title, number, and date of this specification.
b. Size of refrigerator required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1 and 4.3 and 6.3).
e. When wood preservative is other than specified (see 3.3.3)
f. When panel fasteners are other than specified (see 3.3.15.3).
g. When three wrenches are required for Marine Corps acquisition (see 3.3.15.4).
h. When padlocks are other than specified (see 3.3.15.5).
i. When refrigerators do not conform to panel schedule shown on Drawing 5-13-2651 (see 3.4).
j. When conveyor door panels are required; number of conveyor door panels required; and when conveyor door panels are to replace standard wall panels (B),or walk-in door panels (C) (see 3.4.4.1).
k. When partition panels are to be omitted (see 3.4.4.1).
l. When walk-in door panel is to be hinged on left hand side (see 3.4.4.2).

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m. When a ramp is required (see 3.4.6).
n. When Marine Corps data plates are required (see 3.6.2.1).
o. When a crate conforming to MIL-C-104 is required (see 5.2.1.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Part identification number (PIN) structure. The PIN's to be used for items acquired by this specification are created as follows:



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6.5 Refrigeration units. Refrigeration units for use with the refrigerators are panel-mounted refrigeration units covered by MIL-R-13312 (see 6.2). These units are completely self-contained and are mounted on refrigerator panel in the opening provided (see Drawing 5-13-2652, panel D of MIL-R-13312 units). Normal requirements of the refrigeration units shall be as follows:

Size of refrigerator (cubic feet)	Number of refrigeration units
600	1
1200	2
1800	2
4000	4

6.6 Subject term (key word) listing.

Kitchen equipment
Preservation of food
Storage, food

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL

Review activities:

Army - AV, MD, ME
DLA - GS

Preparing activity:

Army - GL

Project No. 4110-0484

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-R-0010932P(GL)	94-08-17

3. DOCUMENT TITLE: REFRIGERATORS, PREFABRICATED, PANEL TYPE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)
U.S. Army Natick RD&E Center	(1) Commercial: 508-651-5175 (2) AUTOVON/DSN: 256-5175
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WEE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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